

Cold-Junction Compensation and Calibration in Thermocouple Modules

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Abstract

This paper presents a comprehensive analysis of cold-junction compensation (CJC) errors in thermocouple measurement modules and evaluates calibration methodologies to mitigate them. Although the AFEA-TC-08 module is utilized as a case study, the underlying physics and proposed calibration techniques are universally applicable to standard thermocouple data acquisition systems.

Contents

1 Principles of Thermocouple Operation	2
2 The Necessity of Cold-Junction Compensation	2
3 The Measurement and Compensation Chain	2
4 Voltage-to-Temperature Conversion and Reference Tables	3
5 Reference Junction Normalization (0 °C)	4
6 Module Firmware Calculations and Error Introduction	5
7 Error Propagation and the Seebeck Coefficient	5
8 Sources of Thermal Gradient Error	6
9 Classification of Measurement Uncertainties	7
9.1 Inherent Thermocouple Bias	7
9.2 Reference Standard Uncertainty	7
10 Evaluation of Calibration Methodologies	7
10.1 Method A: Single-Point Offset Calibration	7
10.2 Method B: Two-Point (Gain and Offset) Calibration	8
10.3 Method C: Permanent Ice-Point Reference Channel (Live CJC)	8
10.4 Method D: Statistical CJC Fitting (Multivariate Regression)	9
10.5 Method E: Direct Terminal Isothermal Sensing (External Sensor)	9
10.6 Method F: Isothermal Terminal Block (Hardware Modification)	10
11 Practical Considerations in Environmental Stability	10

12 Recommended Calibration Protocol	10
13 Procedural Guidelines	11
14 Validation and Limitations	12
Appendix: Quick Symbol Reference	13

1 Principles of Thermocouple Operation

A thermocouple consists of two dissimilar metal wires joined at one end (the measurement or hot junction). A temperature gradient along the wires generates a small electromotive force (EMF) across the open ends—typically a few tens of microvolts per degree Celsius. This phenomenon, known as the Seebeck effect, forms the fundamental principle of thermocouple thermometry.

Crucially, a thermocouple does not measure absolute temperature; it measures a temperature differential. For instance, placing a measurement junction in a 100°C environment while the reference ends remain at ambient temperature yields a voltage dependent on both temperatures, not solely the elevated one. This differential nature is the primary factor driving compensation requirements.

2 The Necessity of Cold-Junction Compensation

Because a thermocouple measures a temperature differential, determining the temperature at the measurement junction requires independent knowledge of the reference junction’s temperature.

This reference junction forms inevitably: a measurement module’s internal circuitry typically consists of copper traces, whereas the thermocouple utilizes specific alloy wires. The physical transition from the thermocouple alloy to the copper terminal creates an unintentional secondary junction. Thus, every practical thermocouple measurement involves two distinct junctions: the deliberate measurement junction (T_{hot}) and the parasitic terminal junction (T_{cold} or T_{terminal}).

To recover the absolute temperature at the primary measurement site, the data acquisition module must accurately ascertain the temperature at the terminal. This process is termed **cold-junction compensation** (CJC).

3 The Measurement and Compensation Chain

Regardless of cost or complexity, standard thermocouple modules execute a four-step sequence:

1. **Differential Voltage Measurement:** The instrument amplifies and digitizes the minuscule differential voltage (V_{meas}) produced by the thermocouple pair.
2. **Terminal Temperature Measurement:** An independent, localized sensor (e.g., a thermistor, an RTD, or a temperature IC) measures the ambient terminal temperature (T_{sensor}).
3. **Voltage Conversion:** The system converts T_{sensor} into an equivalent reference voltage (V_{CJC}) using standardized reference tables for the specific thermocouple type.
4. **Signal Summation:** The module adds V_{meas} and V_{CJC} , then applies the inverse reference table to resolve the final temperature (T_{reported}).

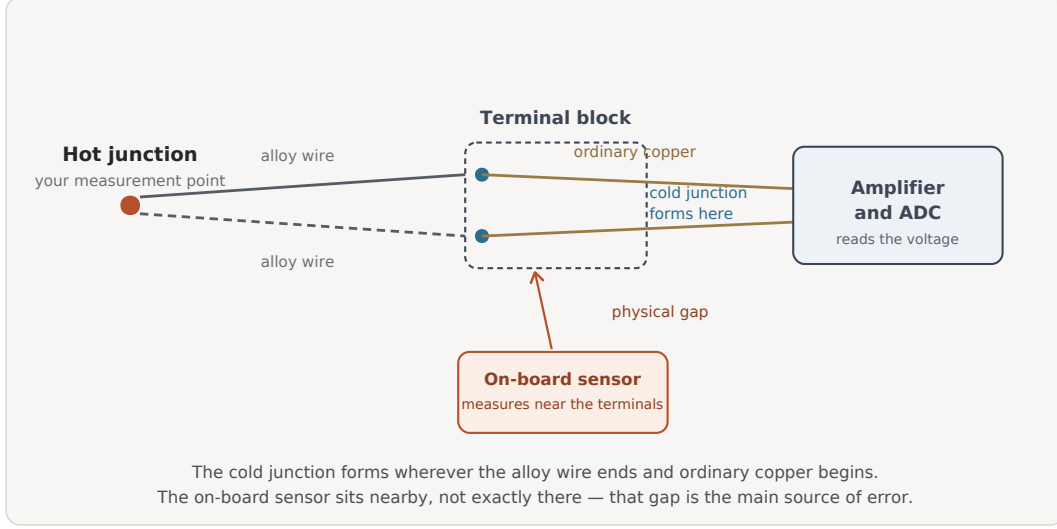


Figure 1: Schematic representation of thermocouple junctions. The cold junction forms where the specific alloy wire terminates at the copper connector. The on-board sensing integrated circuit (IC) is located nearby, and the resulting thermal gradient between these two points constitutes the primary source of CJC error.

Steps 1, 3, and 4 are constrained by physical laws and international metrology standards. Step 2 represents the primary divergence in hardware design and predominantly dictates the module’s real-world accuracy, often overshadowing ADC resolution limitations.

Cold-Junction Architecture	Typical Accuracy	Common Application
PCB-mounted IC displaced from terminals	± 0.5 to ± 2.0 °C	Low-cost modules (e.g., AFEA-TC-08)
IC positioned directly beneath terminal block	± 0.3 to ± 0.5 °C	Mid-range industrial modules
Embedded RTD in a thermally isolated block	± 0.1 °C	Laboratory-grade acquisition systems
External ice-point reference cell	± 0.01 °C	Metrology-lab standards

Table 1: Comparison of cold-junction sensing architectures and their typical baseline accuracies.

The AFEA-TC-08 features an instrumentation-amplifier front end, a 16-bit simultaneous-sampling ADC, and a PCB-mounted temperature sensor for CJC. Although the sensor itself is specified at roughly ± 0.3 °C, its effective contribution to measurement uncertainty is typically higher. The limiting factor is not the sensor’s inherent precision, but the degree to which its reading diverges from the actual temperature at the physical terminal block.

4 Voltage-to-Temperature Conversion and Reference Tables

Converting a thermocouple’s voltage into a temperature requires a standardized reference function. Metrology bodies publish these tables, denoting the EMF generated when the reference junction is held at exactly 0 °C and the measurement junction is at temperature T . This paper designates the

reference EMF function as $E(T)$ and its inverse relation as $E^{-1}(V)$.

Table 2 provides selected reference values for a Type E thermocouple, utilized as the standard example throughout this analysis.

T	$E(T)$
$-100\text{ }^{\circ}\text{C}$	-5.237 mV
$0\text{ }^{\circ}\text{C}$	0.000 mV
$25\text{ }^{\circ}\text{C}$	1.495 mV
$100\text{ }^{\circ}\text{C}$	6.319 mV
$200\text{ }^{\circ}\text{C}$	13.421 mV

Table 2: Selected standardized EMF reference values for Type E thermocouples.

While module firmware inherently manages these nonlinear lookups, a clear conceptual understanding of $E(T)$ is crucial for rigorous error analysis.

Data Integrity Note The EMF values referenced herein are illustrative approximations. Formal calibration procedures must reference the exact NIST ITS-90 thermocouple databases, as typographical errors in reference arrays are exceptionally difficult to isolate from resultant telemetry.

5 Reference Junction Normalization ($0\text{ }^{\circ}\text{C}$)

A common source of confusion is that neither junction in a typical measurement setup resides at $0\text{ }^{\circ}\text{C}$. If a probe measures $100\text{ }^{\circ}\text{C}$ and the terminal block rests at $23\text{ }^{\circ}\text{C}$, the $0\text{ }^{\circ}\text{C}$ -based table must still be utilized.

The $0\text{ }^{\circ}\text{C}$ reference acts as a standardizing mathematical anchor, analogous to Coordinated Universal Time (UTC) for time zones. Just as the temporal difference between two cities is calculated by comparing their offsets from UTC, the voltage between any two temperatures is calculated by routing the differential through $0\text{ }^{\circ}\text{C}$:

$$V_{\text{meas}} = E(T_{\text{hot}}) - E(T_{\text{terminal}})$$

Numerical Example: For a terminal at $23.20\text{ }^{\circ}\text{C}$ and a probe at $100.00\text{ }^{\circ}\text{C}$:

$$\begin{aligned} V_{\text{meas}} &= E(100.00\text{ }^{\circ}\text{C}) - E(23.20\text{ }^{\circ}\text{C}) \\ &= 6.319\text{ mV} - 1.386\text{ mV} \\ &= 4.933\text{ mV} \end{aligned}$$

The ADC only registers this 4.933 mV signal. Consequently, if $T_{\text{hot}} = T_{\text{terminal}}$, the terms cancel perfectly and $V_{\text{meas}} = 0$. This can be empirically observed: if a probe and module achieve thermal equilibrium in the same environment, the channel reports the on-board sensor reading, as the thermocouple itself contributes zero differential voltage.

Assumption of Wire Homogeneity The additive property relies on the assumption that the thermocouple wire remains metallurgically homogeneous along its entire length. Work-hardening, sharp kinking, or severe thermal gradients after long-term aging can introduce microscopic compositional variations, generating localized parasitic EMFs. Algorithmic calibration cannot detect these anomalies, necessitating proper physical handling and periodic physical re-verification of probes.

6 Module Firmware Calculations and Error Introduction

The instrument receives V_{meas} . To calculate T_{hot} , the exact terminal temperature (T_{terminal}) is required. However, the system relies on the on-board sensor reading (T_{sensor}), which serves as an imperfect proxy. The final reported temperature (T_{reported}) is calculated as:

$$T_{\text{reported}} = E^{-1}(V_{\text{meas}} + E(T_{\text{sensor}}))$$

Numerical Example: If the on-board sensor erroneously reports 25.00 °C instead of the true 23.20 °C terminal temperature:

Parameter	Symbol	Value
Differential voltage generated	V_{meas}	4.933 mV
On-board sensor reading	T_{sensor}	25.00 °C
Equivalent CJC voltage	$E(25.00\text{ °C})$	1.495 mV
Total compensated voltage	V_{total}	6.428 mV
Final reported temperature	$E^{-1}(6.428\text{ mV})$	101.62 °C

Table 3: Propagation of CJC error through standard firmware operations.

Despite the probe residing at exactly 100.00 °C, the system reports 101.62 °C. The 1.80 °C error at the cold junction manifests as a 1.62 °C error in the final reading.

From this point forward, the thermal gradient between the sensor and the true terminal is defined as the primary CJC error (ΔT):

$$\Delta T = T_{\text{sensor}} - T_{\text{terminal}}$$

7 Error Propagation and the Seebeck Coefficient

The error ΔT does not propagate at a strict 1:1 ratio because the reference curve $E(T)$ is non-linear. The localized slope of this curve is the Seebeck coefficient, defined as $S(T) = \frac{dE}{dT}$ (typically expressed in $\mu\text{V}/^\circ\text{C}$).

Because the error is injected at the cold junction’s thermal slope but mathematically resolved at the hot junction’s slope, the propagated temperature error (E_{error}) is scaled by the ratio of these coefficients:

$$E_{\text{error}} \approx \Delta T \left[\frac{S(T_{\text{terminal}})}{S(T_{\text{hot}})} \right]$$

For a Type E thermocouple, this ratio remains near unity across a broad range but clearly attenuates at high temperatures:

T_{hot}	$S(T)$ ($\mu\text{V}/^\circ\text{C}$)	Ratio $S(25)/S(T)$
0 $^\circ\text{C}$	58.7	1.04
25 $^\circ\text{C}$	61.0	1.00
100 $^\circ\text{C}$	67.5	0.90
200 $^\circ\text{C}$	74.5	0.82
300 $^\circ\text{C}$	78.0	0.78
500 $^\circ\text{C}$	81.0	0.75

Table 4: Type E Seebeck coefficients and resulting error propagation scaling ratios.

Validating the previous example: $1.80 \times \frac{61.0}{67.5} = 1.62^\circ\text{C}$, aligning perfectly with the E^{-1} inversion.

Two critical behaviors emerge from this relationship. First, a positive error in the cold junction always induces a positive error across all channels. Second, when measuring targets colder than the terminal block, the ratio exceeds 1.0, effectively amplifying the CJC error due to the reduction of $S(T)$ at lower temperatures.

8 Sources of Thermal Gradient Error

Several factors contribute to the mismatch between T_{sensor} and T_{terminal} , presented in general order of magnitude:

1. **Spatial Displacement:** The on-board sensor is typically surface-mounted on the PCB, separate from the terminal block. Any lateral thermal gradient across the substrate introduces direct error.
2. **Internal Power Dissipation:** Active components (amplifiers, ADCs, voltage regulators) dissipate heat. Consequently, the internal PCB geometry operates at an elevated temperature compared to the exposed edge terminals, which also experience heat-sinking effects from connected thermocouple wires.
3. **Intra-Block Gradients:** In multi-channel systems, terminals at opposite ends of a connector block rarely maintain identical temperatures. A singular centralized sensor cannot accurately represent both extrema.
4. **Thermal Transients:** The PCB and terminal block possess disparate thermal masses, resulting in divergent heating rates. Immediately following power-up or a sudden shift in ambient conditions, ΔT spikes significantly compared to steady-state operations.
5. **Environmental Convection:** Cooling fans, open enclosures, or localized drafts alter convection profiles. A module characterized on an open laboratory bench will exhibit different error characteristics once installed in a sealed chassis.

Static contributors (1-3) can be largely mitigated via steady-state calibration. Dynamic factors (4-5) require environmental control or continuous live-reference correction.

9 Classification of Measurement Uncertainties

To evaluate calibration methodologies objectively, measurement errors must be categorized by their mathematical behavior:

- **Systematic Error:** A repeatable bias occurring under identical conditions. Statistical fitting effectively removes systematic error.
- **Random Noise:** High-frequency variance with no fixed pattern. Calibration cannot remove noise; it can only be attenuated via oversampling and digital filtering.
- **Temporal Drift:** A systematic bias that evolves over time (e.g., metallurgical aging, terminal oxidation). Static calibration nullifies drift instantaneously, but the model degrades as time progresses.

While CJC offset (ΔT) constitutes a major systematic error, it is not the sole variable.

9.1 Inherent Thermocouple Bias

Physical thermocouple wire deviates slightly from the idealized reference table $E(T)$ due to manufacturing variances. This bias can be expressed as a function of temperature:

$$E_{\text{wire}}(T) = T_{\text{true}} - T_{\text{table}}$$

For undamaged wire, $E_{\text{wire}}(T)$ is highly repeatable (systematic). A standard two-point calibration models the linear component of this bias:

$$E_{\text{wire}}(T) \approx \beta_0 + \beta_1 T$$

Solving for the intercept (β_0) and slope (β_1) nullifies the linear deviation. For standard base-metal thermocouples operating over moderate ranges, the residual non-linearity is typically well below raw manufacturing tolerances (often resolving to tenths of a degree rather than the standard $\pm 1.5^\circ\text{C}$ Class-1 spec).

9.2 Reference Standard Uncertainty

Calibration efficacy is strictly bounded by the accuracy of the reference standard. An improperly mixed ice bath or an uncorrected boiling point establishes a fundamental error floor. Calibrations cannot yield absolute accuracies exceeding the certainty of the applied reference nodes.

10 Evaluation of Calibration Methodologies

Calibration algorithms map raw module output against known thermodynamic states to extract corrective coefficients. The following methodologies range from static offset corrections to dynamic hardware interventions.

10.1 Method A: Single-Point Offset Calibration

Principle: Submerge the probe in a single known reference state (e.g., 0°C). The differential between the reported value and the standard represents the combined systematic error (wire bias + CJC offset) at that specific thermal state.

Calculation:

$$O_{\text{new}} = O_{\text{old}} + (T_{\text{ref}} - \bar{T}_{\text{reported}})$$

Advantages	Disadvantages
Minimal procedural overhead (requires one bath). Simultaneously mitigates wire bias and CJC error.	Valid only near the specific calibration point. Incapable of correcting slope/gain variations.
Hardware-agnostic application.	Vulnerable to subsequent thermal baseline shifts.

10.2 Method B: Two-Point (Gain and Offset) Calibration

Principle: Two thermal reference nodes generate a system of equations, isolating both slope (G) and intercept (O). The calculated slope inherently absorbs the Seebeck attenuation ratio and the linear portion of the wire bias.

Calculation: Given normalized readings u_1 and u_2 derived from previous calibration states:

$$u_1 = \frac{T_1 - O_{\text{old}}}{G_{\text{old}}}$$

$$u_2 = \frac{T_2 - O_{\text{old}}}{G_{\text{old}}}$$

$$G_{\text{new}} = \frac{T_{\text{ref},2} - T_{\text{ref},1}}{u_2 - u_1}$$

$$O_{\text{new}} = T_{\text{ref},1} - u_1 G_{\text{new}}$$

Advantages	Disadvantages
Compensates across a wide operational temperature range.	Requires careful maintenance of two distinct thermal baths.
Absorbs both signal attenuation and inherent wire slope error.	Relies on a static instrument thermal state during calculation.
Considered the industry-standard baseline methodology.	Requires strict barometric/altitude correction for boiling points.

10.3 Method C: Permanent Ice-Point Reference Channel (Live CJC)

Principle: Dedicate one multiplexed channel to a permanent 0°C ice bath. Since $E(0) = 0$, the reported temperature on this reference channel isolates the module's live CJC error.

Calculation:

$$T_{\text{corrected}} = T_{\text{reported}} - \left(\bar{T}_{\text{ref_channel}} \left[\frac{S(T_{\text{terminal}})}{S(T_{\text{hot}})} \right] \right)$$

Advantages	Disadvantages
Actively tracks warm-up phases and real-time environmental drift.	Reduces available data channels by one.
Eliminates the assumption of steady-state thermal gradients.	Requires continuous maintenance of a physical 0 °C reference.
Provides direct empirical measurement of CJC variance.	Introduces the reference probe’s own minor tolerance error.

10.4 Method D: Statistical CJC Fitting (Multivariate Regression)

Principle: Model the true temperature analytically by incorporating the on-board CJC sensor telemetry directly into a multi-variable regression, circumventing the need for hardware additions.

$$T_{\text{true}} = \beta_0 + \beta_1 T_{\text{reported}} + \beta_2 T_{\text{sensor}}$$

Collinearity Constraints: The coefficient β_2 is mathematically indeterminate if the module’s internal temperature remains static during data acquisition. In steady-state, T_{sensor} becomes highly collinear with the intercept β_0 , yielding statistically meaningless noise for β_2 . To ensure parameter identifiability, the system must undergo forced thermal transitions (e.g., sampling continuously from cold-boot through a 40-minute warm-up) during the calibration sequence.

For extensive datasets ($n \geq 5$ points), polynomial expansion enhances accuracy by characterizing non-linear wire curvature:

$$T_{\text{true}} = \beta_0 + \beta_1 T_{\text{reported}} + \beta_2 T_{\text{reported}}^2 + \beta_3 T_{\text{sensor}}$$

Advantages	Disadvantages
Purely empirical characterization; circumvents theoretical Seebeck ratios.	Demands rigorously structured, transient data collection protocols.
Provides dynamic drift mitigation via software alone.	Yields invalid models if internal thermal variance is insufficient.
Yields analytical insight into internal firmware efficacy (if $\beta_2 \approx 1$).	Confounded if the ambient environment tracks perfectly with self-heating.

10.5 Method E: Direct Terminal Isothermal Sensing (External Sensor)

Principle: Physically bypass the imprecise on-board CJC sensor by bonding a high-precision reference thermometer (e.g., a calibrated PT100 RTD) directly to the terminal block.

Calculation: First, mathematically strip the module’s internal CJC addition:

$$V_{\text{meas}} = E(T_{\text{reported}}) - E(T_{\text{sensor_internal}})$$

Subsequently, recalculate absolute temperature using the external high-precision telemetry:

$$T_{\text{true}} = E^{-1}(V_{\text{meas}} + E(T_{\text{sensor_external}}))$$

Advantages	Disadvantages
Rectifies the physical source of error (spatial displacement).	Necessitates custom hardware modification and dual-acquisition pipelines.
Preserves full channel density without requiring ice baths.	Accuracy is heavily dependent on the thermal interface material bond.

10.6 Method F: Isothermal Terminal Block (Hardware Modification)

Principle: Unlike algorithmic corrections, this hardware approach minimizes intra-block gradients. Enclosing the terminal block in high-mass aluminum or thermally conductive potting compound forces spatial thermal equilibrium across all channels and the sensor.

Advantages	Disadvantages
Physically eliminates channel-to-channel baseline discrepancies.	Requires permanent, invasive mechanical modification to the PCB/chassis.
Operates synergistically alongside any algorithmic calibration method.	Substantially increases the system’s thermal settling time constant.

11 Practical Considerations in Environmental Stability

Operational concerns frequently focus on absolute module temperature and ambient fluctuations. An analysis of the thermodynamics reveals where calibration efforts are most impactful.

Absolute instrument temperature is non-critical. The error term is inherently differential (ΔT). If internal power dissipation uniformly elevates the PCB by 15 °C, both the on-board sensor and the terminal block temperatures rise identically. A module operating stably at 40 °C demonstrates equivalent accuracy to one operating at 25 °C.

Uniform ambient shifts are largely benign. An ambient rise of 2 °C uniformly shifts the assembly, preserving ΔT . The residual errors resulting from this shift (minor alterations in convection coefficients and shifts in localized Seebeck properties) total less than 0.05 °C, rendering them negligible compared to wire tolerances.

Localized gradient shifts are the primary vulnerability. Directional airflow (e.g., HVAC draft), localized solar loading, the actuation of adjacent high-current relays, or the physical insertion of probes (which act as thermal shunts) alter the spatial gradient between the sensor and terminals. Therefore, it is mathematically imperative to characterize the system in its final deployed configuration.

12 Recommended Calibration Protocol

To achieve optimal accuracy in industrial and laboratory environments, the following procedure is recommended:

1. **Firmware Configuration:** Configure the appropriate thermocouple profile within the hardware to ensure the baseline $E(T)$ linearization matrix is correct.
2. **Data Sanitization:** Nullify existing factory or legacy calibration registers to prevent recursive error accumulation.

3. **Thermal Stabilization:** Allow the system to reach thermal steady-state (typically 30–45 minutes). Stabilization must be confirmed by derivative analysis of the CJC telemetry ($dT/dt \approx 0$), rather than arbitrary timing.
4. **Data Acquisition:** Execute a two-point calibration protocol utilizing a 0 °C ice-slush reference and an altitude/pressure-corrected boiling reference.
5. **Metadata Archival:** Record the steady-state CJC reading alongside the calibration coefficients to establish the baseline thermal state.
6. **Validation:** Verify the generated model against an independent intermediate thermal node to ensure linearity and confirm the absence of polynomial distortion.

Implementation via the standard AFEA_TCCopysource framework:

```
tc.setChannelType(1:8, 'E');
tc.resetCalibration(1:8);

% Empirical characterization of internal thermal drift
out = AFEA_warmupDrift(tc, 45);

% Execution of two-point algorithmic calibration
tc.calibrateTwoPoint([0 100], bath1_readings, bath2_readings);

% Optional: Implementation of statistical live drift compensation
tc.CJCComp.Factor = out.betaMean;
tc.CJCComp.Enabled = true;
tc.setCJCReference();
```

13 Procedural Guidelines

Preparation Phase:

- Ensure module thermal stabilization is mathematically confirmed via telemetry.
- Prepare the ice bath using crushed ice and minimal distilled water to form a dense slush; floating ice matrices induce severe thermal stratification.
- Ensure the physical chassis and airflow topology exactly mirror deployment conditions.

Execution Phase:

- Ensure probe immersion depth exceeds 60 mm to negate stem conduction artifacts.
- Compute the local boiling point rigorously, correcting for altitude ($-0.33\text{ °C per }100\text{ m}$) and local barometric pressure.
- Allow a minimum of 5 minutes for probe equilibrium before initiating data capture.
- Average a minimum of 30 discrete samples per thermal node to attenuate Gaussian noise.

Post-Calibration Phase:

- Conduct an intermediate independent temperature validation.
- Perform temporal re-verification at 72 hours to ensure model stability against drift.

14 Validation and Limitations

The theoretical frameworks, algorithms, and representative Type E coefficients detailed in this paper form a robust methodology for CJC analysis. However, exact deployment necessitates the utilization of localized NIST ITS-90 databases for quantitative coefficient extraction.

Furthermore, internal firmware operations (such as the extent of proprietary CJC pre-processing within modules like the AFEA-TC-08) are inferred from empirical observation rather than proprietary schematics. Method D explicitly serves as a diagnostic tool in this regard: an extracted regression coefficient (β_2) approaching unity indicates a lack of internal compensation, dictating that subsequent mitigation must occur entirely within user-space software or via external hardware intervention.

Appendix: Quick Symbol Reference

Symbol	Meaning
T_{hot}	The absolute temperature at the primary measurement junction
T_{terminal}	The true absolute temperature at the physical alloy-to-copper transition
T_{sensor}	The temperature reported by the data acquisition module's internal CJC sensor
T_{reported}	The final, pre-calibrated temperature array output by the module
ΔT	The primary spatial thermal gradient ($T_{\text{sensor}} - T_{\text{terminal}}$)
V_{meas}	The physical differential electromotive force (EMF) generated by the thermocouple
$E(T)$	Standardized EMF function mapping absolute temperature to voltage (referenced to 0 °C)
$E^{-1}(V)$	The inverse standardized function mapping voltage back to absolute temperature
$S(T)$	The Seebeck coefficient; the local derivative $\frac{dE}{dT}$, typically expressed in $\mu\text{V}/^\circ\text{C}$